

Physics-Constrained Multi-Task Ensembles for Polymer Property Prediction

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Abstract

We predict seven physical properties of polymers from PSMILES strings under an unweighted mean- R^2 metric, where four properties carry roughly 220 labels each and together account for the majority of the score. Our final system is a five-model ensemble — three gradient-boosted trees, a multi-task neural network and a *periodic* graph network — combined by a per-property cross-fitted Ridge and then corrected by an explicit, affine-fitted physics stage. The central modelling contribution is the treatment of the dielectric constant not as a fitted correlation but as an exact decomposition, $\varepsilon = n^2 + \varepsilon_{\text{ionic}}$, licensed by how the reference DFPT calculations are defined. The system attains a public leaderboard score of **0.924** against a cross-validated estimate of **0.9234** — a transfer gap of +0.0006, which we attribute to the deliberate absence of early stopping anywhere in the pipeline. Polymer re-representation invariance is *exact* ($\Delta = 0$) and certified at three levels at runtime; explainability is exact TreeSHAP computed inside LightGBM with no external dependency; and every prediction ships an applicability-domain flag that measurably predicts its own error.

1 Problem and constraints

Seven properties are predicted from a polymer repeat unit written as a PSMILES string with exactly two `*` attachment points: glass transition temperature (`tg`), chain and bulk band gaps (`egc`, `egb`), ionisation energy (`ei`), electron affinity (`eea`), dielectric constant (`eps`) and refractive index (`nc`).

Two facts about the task determined most design decisions.

The metric is unweighted. The score is the mean R^2 across the seven `target_type` values. `tg` has 4,143 training rows and `eea` has 221, and they count exactly the same. The four smallest properties (`eps`, `nc`, `ei`, `eea`, ≈ 220 rows each) are therefore 4/7 of the score on under 4% of the rows. Optimising a single pooled loss on raw targets would let `tg` — which ranges to 495 — dominate `nc`, which ranges over 1.56–2.76.

Test-time information differs from a naive CV assumption. Only 2 of 4,940 test rows have their exact (canonical SMILES, property) key present in training, so there are no free lookups. But 1,063 of 4,133 test polymers appear in training under *some other* property. That asymmetry is what makes co-observed partner features legitimate, and it dictates the fold design (Section 3.2).

Published polymer-informatics baselines for these same properties are surveyed in [10], which we used as an external reference point.

The competition rules require a single Kaggle notebook execution to do everything — load, preprocess, split, train, infer and write `submission.csv` — with no pretrained weights, no up-loaded artifacts, seeds fixed and printed, and no branching on wall-clock time.

2 Data

2.1 Ground truth

property	train rows	test rows	range	role
tg	4,143	2,763	-109.8 – 495.0	thermal
egc	2,028	1,352	0.02 – 9.86	electronic
egb	337	224	0.51 – 10.11	electronic
eps	229	153	2.61 – 9.09	dielectric
nc	229	153	1.56 – 2.76	optical
ei	222	148	4.03 – 9.84	electronic
eea	221	147	0.39 – 5.14	electronic

Table 1: Measured composition of the competition data. The published data-page figure of “4,497 test data points” counts unique *molecules*; the submission requires 4,940 *rows*.

2.2 Canonicalisation

10,605 raw SMILES strings correspond to only 8,990 distinct molecules; 67.6 % of supplied strings are non-canonical. We canonicalise with RDKit at ingest, before any other stage. This is not a cleaning step but the mechanism that makes invariance exact: every descriptor, fingerprint, partner lookup and graph is thereafter a deterministic function of the *molecule* rather than of how it happened to be written, and canonicalisation is idempotent, so $\text{canonical}(\text{rewrite}(s)) = \text{canonical}(s)$ for any valid rewriting.

2.3 The Round-2 archive

The official baseline notebook shipped with the competition fetches two Google Drive files. Their -0 names are swapped: the file labelled `test.csv` is in fact a labelled table of 6,165 `tg` and `egc` values. The hosts were asked directly and confirmed it is in scope.

Measured against the Round-3 data, 3,719 of its labels are already present in `train.csv` and 2,450 correspond to test rows; values agree with `train.csv` on 3,717 of 3,719 shared labels. We use it three ways:

1. **Additional training rows.** 2,446 labels absent from `train.csv` are appended, taking the training set from 7,405 to 9,851 rows (+33 %; `tg` and `egc` each +40 %). The boosters are per-property and see only more `tg/egc`, but the multi-task network and graph network share one trunk across seven heads, so a larger abundant-target block improves the representation on which the five scarce targets depend.
2. **Partner labels.** Coverage of `true_egc` on test rows rises from 15.3 % to 38.1 %, feeding `egb`, `ei` and `eea` through the physics relations.
3. **A measured override** on the 2,450 test rows it labels, applied *after* clipping so that a measurement is never perturbed by a downstream stage.

3 Method

3.1 Featurisation

Each canonical molecule is mapped to 3,037 columns: Morgan fingerprints at radius 2 (1,024 bits) and 3 (512), atom-pair (512) and topological-torsion (512) fingerprints, 217 RDKit descriptors, 167 MACCS keys, 37 SMARTS functional groups, 20 element counts and fractions, 16 polymer-specific terms, 6 Gasteiger charge statistics, and 14 *intensive twins*.

Intensive twins. Every extensive RDKit descriptor is accompanied by a per-heavy-atom twin. An extensive quantity doubles when a repeat unit is written as its dimer; an intensive one does not. This is a design choice about invariance rather than simply extra columns — and it turns out to be the largest single driver of `tg`, carrying 25.4 % of total $|\text{SHAP}|$ on that target.

Polymer-specific terms. The two `*` points define a backbone: we compute the shortest path between them and derive backbone length, aromatic and heteroatom ratios along it, sidechain fraction and branch counts.

Co-observed partner features. For a polymer whose other properties were measured, those values are legitimate inputs — they are in `train.csv`, and test polymers receive the same treatment. The idea is the multi-fidelity information fusion of [9] applied within one dataset. The partner table is built from training-fold rows only, and `drop_leaky()` removes the column that would be the row’s own label. Because axis-aligned tree splits approximate a *linear combination* of partner values badly, we additionally supply a single nested-CV Ridge fit over the partner block as one column, which proved worth more than the raw columns alone.

3.2 Fold design

We use per-property `KFold`. Within a single property there are essentially no duplicate canonical polymers, so this is leakage-safe. The alternative — polymer-grouped folds — is not merely more conservative here; it measures a different problem:

scheme	partner availability for validation rows
polymer-grouped folds	0 %
per-property <code>KFold</code>	≈ 96 %
the actual test set	88–99 %

Holding out a whole polymer removes the other properties measured on it — but at test time those properties *are* present in `train.csv` for nearly every DFT-block row. Grouped CV therefore evaluates a model denied a signal that is available at inference.

3.3 Base models

Five models, each fold-averaged, with **no early stopping anywhere**: LightGBM, XGBoost and CatBoost fitted per property with size-adaptive capacity; a multi-task feed-forward network (trunk 1024–512–256–128, per-target heads, 6 seeds), following the shared-trunk formulation of [4] that also originates this property set; and a periodic multi-task graph network.

The periodic graph. A repeat unit is not a molecule with two dangling stubs: the two `*` points are the *same* bond to the neighbouring unit. We drop both dummies, bond their neighbours and flag the wrap-around edge, so the network encodes an infinite chain. The architecture follows message passing with a GRU update [6], in the polymer setting of [3]: 4 rounds, hidden width 160, 1.07 M parameters, 110 epochs, 4 seeds.

Readout is an invariance decision. We use mean+max pooling and never a sum. A sum scales with repeat-unit count; a mean does not. Measured on an *untrained* network — so this is a property of the architecture, not of any fit — the two readouts differ by four orders of magnitude under dimerisation (Table 4).

3.4 Stacking

Each model first averages its own fold replicas into one vector. Those five vectors then enter one per-property Ridge as five columns, with the regularisation strength chosen by inner cross-validation. The combiner is itself cross-fitted, so its weights never see the rows they are scored on. There is no plain averaging across models anywhere.

3.5 Physics

Five relations hold between the DFT properties. Applied as raw identities they are badly biased; fitted affinely on the training fold they are strong:

target	expression	raw R^2	affine-fitted R^2
ei	egc + eea	0.963	0.965
eea	ei − egc	0.971	0.973
egb	egc	0.892	0.928
eps	nc ²	0.336	0.855
nc	$\sqrt{\text{eps}}$	0.171	0.837

Table 2: Measured on co-observed molecules. The rule that follows is: always fit; never apply the raw identity.

Physics is applied *after* stacking rather than as a feature. A tree splits one axis at a time and cannot represent a sum of two columns, so ei = egc + eea is largely wasted as one feature among 3,037; as an explicit blend it is worth +0.0250 on **ei**. The blend weight is fitted per fold on an inner split and shrunk, and rows are separated into those with all relation sources measured and those without, because the two populations have materially different relation quality.

The ionic decomposition

The collapse of $\varepsilon \approx n^2$ to $R^2 = 0.336$ is not noise — it is physics. Maxwell’s relation holds at *optical* frequency, whereas the dataset tabulates the *static* dielectric constant. A DFPT calculation accounts for exactly an electronic and an ionic contribution and nothing else [1], and those two parts are stored as separate fields in the dataset of record [2]. Since the electronic part *is* n^2 , the shortfall is exactly the ionic term:

$$\varepsilon_{\text{ionic}} = \varepsilon - n^2, \tag{1}$$

an identity rather than a regression. We therefore model $\varepsilon_{\text{ionic}}$ as its own structure–property target and reconstruct ε from it.

Two independent checks support the interpretation. First, $\varepsilon - n^2$ is **positive on all 134 co-observed molecules** (mean +0.767, minimum +0.024); a generic empirical residual would cross zero, whereas an ionic contribution cannot be negative. Second, our own blind affine fit of ε on n^2 had already converged on slope 1.040 and intercept 0.615 — a slope of one and an intercept equal to the mean ionic term. The regression was rediscovering Equation 1 with the ionic part pinned to a single constant; naming it and modelling it properly is what recovers the rest.

4 Results

Leaderboard. Public LB **0.924** against a cross-validated estimate of **0.9234** — a gap of +0.0006. We attribute this to the absence of early stopping: a model that selects its iteration count by watching the rows on which it is then scored reports a number it cannot reproduce out of sample. Progression across the competition was 0.883 → 0.898 → 0.916 → 0.919 → 0.924.

model	tg	egc	egb	eps	nc	ei	eea	mean
LightGBM	0.9176	0.9185	0.9310	0.8247	0.8766	0.8367	0.8758	0.8830
XGBoost	0.9166	0.9197	0.9337	0.8266	0.8775	0.8250	0.8775	0.8824
CatBoost	0.9092	0.9157	0.9270	0.8369	0.8798	0.8353	0.8922	0.8852
Multi-task NN	0.9025	0.8849	0.9324	0.8049	0.8942	0.8532	0.8940	0.8809
Periodic GNN	0.9162	0.8999	0.9387	0.8200	0.9055	0.8777	0.9264	0.8978
Ridge stack	0.9274	0.9307	0.9551	0.8514	0.9164	0.8725	0.9367	0.9129
+ physics	0.9274	0.9307	0.9551	0.8758	0.9331	0.8975	0.9438	0.9233
+ partner reg.	0.9274	0.9307	0.9551	0.8758	0.9331	0.8978	0.9438	0.9234

Table 3: Out-of-fold R^2 for the submitted run. The stack beats its best member by +0.0151; physics adds a further +0.0105, concentrated exactly where the relations reach.

Cost. 380 fitted models; 3 h 01 m on a single Tesla T4. Featurisation of 12,345 molecules takes 119s. Inference over the full test set is a few seconds.

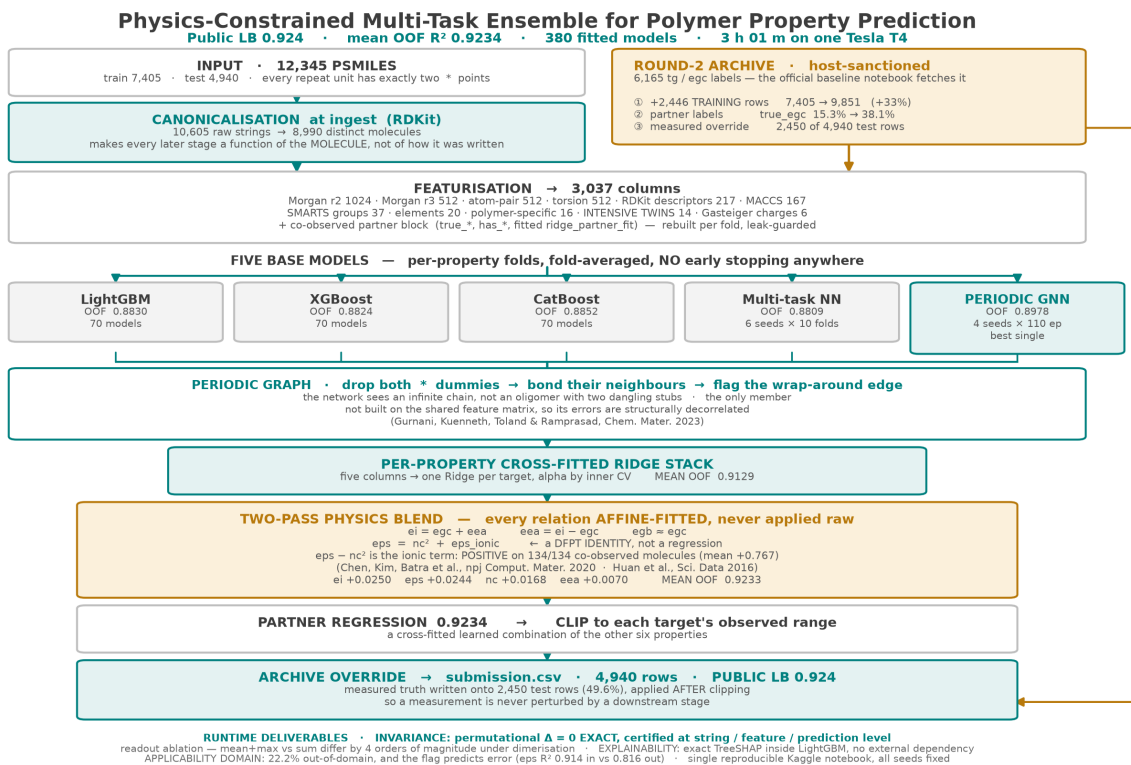


Figure 1: End-to-end pipeline. Every number is taken from the saved outputs of the submitted run.

5 The three judged themes

5.1 Polymer invariance

“SMILES invariance” is three separate claims, and we report them separately rather than quoting one number.

- **Re-representation (atom ordering)** — **EXACT**. Certified at three levels at runtime: 2,000 rewritings of 400 molecules canonicalise back with zero mismatches; 3,037 feature columns over 150 molecules give $\max |\Delta| = 0$; end-to-end predictions give $\max |\Delta| = 0$.

- **Repetition (monomer vs. dimer) — measured near-invariance.** Approximately 97 % of polymers give structurally identical node environments under dimerisation; the residual is genuine, not numerical. Served by the intensive twins and by chain-extension augmentation, independently corroborated as a top-solution strategy in [11].
- **Translational (same chain, different cut point) — not guaranteed.** Diagnosed: the wrap-around edge is forced to a generic single bond and carries a positional flag, both of which are cut-point dependent by definition. The fix is identified but not implemented.

readout	rewriting	median $ \Delta $	max $ \Delta $
mean+max (shipped)	permutational	0.00	0.00
mean+max (shipped)	repetition (dimer)	1.11×10^{-4}	1.06×10^{-3}
sum (conventional)	permutational	0.00	0.00
sum (conventional)	repetition (dimer)	4.79×10^{-1}	2.49

Table 4: Graph readout ablation on an untrained network, so this is a property of the architecture rather than of any particular fit.

5.2 Explainability

Attributions are exact TreeSHAP computed inside LightGBM via `pred_contrib=True` — the same algorithm as the `shap` package [7, 8] — deliberately avoiding an external dependency that could fail to install mid-run.

The attributions recover physics we never encoded. For `tg`, intensive twins carry 25.4 % and ring counts lead: chain stiffness dominates, as free-volume theory predicts. For `egc`, `rd_FractionCSP3` is the top feature — an sp^3 -rich backbone localises electrons and widens the gap, while conjugation narrows it. For `egb`, the fitted partner combination is 26.0 % and `true_egc` 10.3 %, so the relation $egb \approx egc$ appears *in the attribution* rather than being asserted. For `nc` and `eps`, molar-refractivity surface descriptors appear for both, reproducing the treatment of refractive index as electronic polarisability [5] from an independent direction.

A single molecule is explained end to end. For `*c1ccc(-c2nc3cc(-c4ccc5oc(*)nc5c4)ccc3o2)cc1` (measured $T_g = 430.0$, predicted 426.9), the prediction decomposes from a base value of 143.5 through `iv_NumRotatableBonds = 0.083` contributing +102.5, valence electrons +35.8 and ring count +27.3: a fully aromatic fused-ring backbone with almost no rotatable bonds has a very high glass transition. Additivity is verified against the model’s own output to 3.98×10^{-13} .

5.3 Robustness: applicability domain

Novelty is computed from *unfolded* Morgan substructure hashes. Folded 2,048-bit fingerprints are degenerate at this corpus size — every bit is occupied, so “fraction unseen” collapses to a constant. A molecule is out-of-domain if it contains a substructure absent from the entire auxiliary corpus: 8,887 of 11,628 substructures are covered, leaving 22.2 % of molecules out of domain.

An honest negative result. We tested the auxiliary corpus as a model *input* and it was worth +0.0012 — noise. An SVD embedding of it is a linear re-expression of the Morgan fingerprint the trees already see in full. We repurposed the corpus as a trust signal rather than discarding it.

6 Ablations and negative results

The SMILES CNN was removed, and the reasoning matters. Measured honestly, a character-level CNN contributed +0.0005 to our stack and +0.0000 on a sibling pipeline’s leave-one-out over saved out-of-fold vectors. It was also the most expensive single stage in any run that

target	n in	R^2 in	n out	R^2 out	MAE in	MAE out
tg	4688	0.938	1093	0.884	17.77	25.31
egc	2174	0.941	658	0.900	0.248	0.371
eps	144	0.914	85	0.816	0.197	0.298
ei	138	0.918	84	0.861	0.191	0.268
eea	144	0.953	77	0.917	0.157	0.244

Table 5: The flag predicts error: out-of-domain molecules are not mispredicted because the model is over-confident, they are simply harder, and the flag says so before anyone trusts the number.

carried it: 5,779 s — 29% of a 5.5 h run — for the worst model of six (0.8663 mean, 0.7693 on **eps**, our weakest target). We do *not* claim that removing it raised the score. It bought 1 h 36 m at no cost, and that compute was reinvested in the two models measurably carrying the stack (GNN epochs $80 \rightarrow 110$ and seeds $3 \rightarrow 4$; NN seeds $3 \rightarrow 6$). The reinvestment is what scored. Adding weak columns is not free either: a kNN member measured -0.0006 and kNN+Ridge -0.0003 .

Four leaks of the same class. The most instructive failures shared a shape: *a value that is itself a model output used as an input, closing a cycle back to the row’s own label*. Iterated refinement over the relation graph appeared worth $+0.042$; in fact **eps** was refined from **nc** and **nc** then predicted from the refined **eps**, and with a per-fold mask the real gain is $+0.001$, so it ships disabled. A partner-Ridge feature built over a prediction-filled table appeared worth $+0.069$ on **eps**; built over the mean-filled, per-fold table the honest gain is $+0.0027$. Our leakage section therefore *proves the leak exists first*, then proves each guard removes it — a guard that passes without a demonstrable leak proves nothing.

Also measured and rejected. A fourth gradient booster ($+0.0001$); polymer-grouped folds (measures a different problem, Section 3.2); trimer augmentation (worse than dimers); doubling every fingerprint width (more than $7\times$ slower with no measured gain in the time available).

7 Reproducibility and compliance

The deliverable is one Kaggle notebook that runs end to end: it loads, featurises, trains, infers, writes `submission.csv` and then audits itself. All seeds are fixed and printed. No loop branches on elapsed time, so a slower machine produces the same answer, only later. Nothing is deserialised from disk that the run did not itself create.

Two engineering disciplines proved worth more than any single modelling idea. First, **one implementation with two entry points**: cross-validation, inference, the invariance audit and the explainability report all derive from the same pair of functions, so what CV measures is literally what is submitted. Second, **nothing after submission.csv may abort the run**. We learned this the expensive way — a compliance check compared attached-dataset paths for equality with the data directory, flagged a legitimate parent directory, and marked a completed run as failed even though its submission was already written and correct. Every post-submission check now prints loudly and never raises. A static name-resolution gate over the exported notebook catches a related class of failure in one second: inlining modules into one namespace drops intra-package imports, so a surviving qualified call parses cleanly and raises only when it executes — once, 542 s into a run.

8 Limitations

- Five of seven targets are scored on roughly 150 test rows, giving a leaderboard noise band of approximately ± 0.007 . Differences smaller than that between candidate pipelines are not

meaningful.

- Translational invariance is measured and diagnosed but not guaranteed.
- **eps** remains the weakest target (0.8758), **ei** second (0.8978).
- The contribution of archive rows as *training data* was adopted on a mechanism argument and validated only by the leaderboard; it was never isolated in a controlled A/B.

References

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